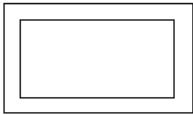
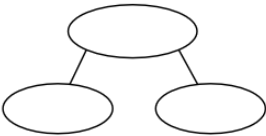
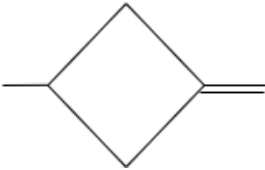
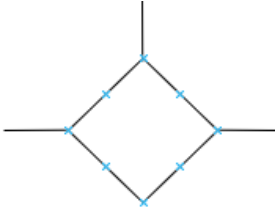
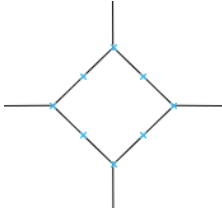
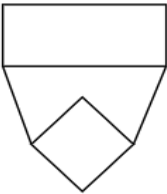
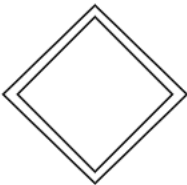
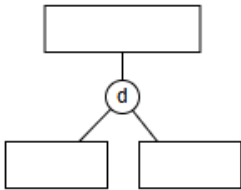
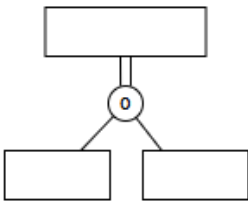


CSE370 Mid Review:

!! Review AFTER you have completed the slides in buX !!

ER/EER

Things to keep in mind:

Strong entity 	Weak entity 	Simple Attributes 	Partial Attribute 
Composite Attribute 	Multivalued Attribute 	Composite + Multivalued 	Key Attribute 
Derived Attribute 	Binary Relationship 	Ternary Relationship 	N-ary Relationship(N>3) 
Recursive Relationship 	Identifying Relationship 	Disjoint Specialization  Disjoint $\Rightarrow$ d Optional Participation $\Rightarrow$	Overlapping Specialization  Overlapping $\Rightarrow$ o Total Participation $\Rightarrow$
<u>Cardinality Ratio</u> 1:1 $\Rightarrow$ One to One, 1:N $\Rightarrow$ One to Many, M:N $\Rightarrow$ Many to Many [Must Use different letters for M:N] M:N:K: Many to Many to Many [Ternary]		<u>Existence Dependency</u> Single Line for Optional Participation Double Line for Total Participation NOTE: If Cardinality and Participation are not Clearly mentioned OR can't be inferred from Question, Make Logical Assumption	

Database Constraints

Things to keep in mind:

- Every Table must have one Primary Key (PK).
- The Primary Key can be composed of one or more attributes.
- The PK is denoted with an underline.
- Foreign Keys are not native to their own table but are referencing (with an arrow) to their native/ referenced table.
- Like this: Foreign Key (FK) —-----> Primary Key (PK)
- Insert, Update operation can cause all 4 constraint violation
- Delete operation can cause only FK constraint

How to identity Database Constraints:

Constraints	Description
1. Key Constraints	No duplicate Primary Key Value
2. Entity Integrity Constraints	No null value for Primary Key
3. Referential Integrity Constrains (FK Constraint)	Inserted new Foreign Key (FK) must match from Referenced Key (PK) (IF the FK is part of PK, else can be null)
4. Domain Constraints	DataType should match

How to enforce Database Constraints:

Constraints	Prevention / Solution / Enforce Constraint
1. Key Constraints	Set the attribute as Primary Key
2. Entity Integrity Constraints	Set the attribute as Primary Key
3. Referential Integrity Constrains (FK Constraint)	On Insert: Set proper Foreign Key with Primary key. On Update/Delete: Restrict → operation does not run Cascade → Delete rows where it matches Set Null → Set Null where it matches
4. Domain Constraints	Set the attribute with proper DataType

Schema Mapping

Things to keep in mind:

Step	Mapping Process	Details
Step 1	Mapping of Regular/Strong Entity Types	Primary Key should be underlined
Step 2	Mapping of Weak Entity Types	Both Partial Key and Foreign Key of Strong Entity should be underlined
Step 3	Mapping of Binary 1:1 Relationship Types	- FK placed in the relation with Total Participation, otherwise any relation is fine. - FK points towards referenced PK (not underlined, use an arrowhead)
Step 4	Mapping of Binary 1:N Relationship Types	- FK placed in the relation on the N side. - FK points towards referenced PK (not underlined, use an arrowhead)
Step 5	Mapping of Binary M:N Relationship Types	- Create a separate relation with both participating entities' Keys as a composite key. - Both keys should be underlined, and both are FKs (they point to their respective PKs).
Step 6	Mapping of Multivalued Attributes	- Create a separate relation with the primary key of that entity and the multivalued attribute. - Both the key and the attribute should be underlined. FK points towards PK.
Step 7	Mapping of N-ary Relationship Types	- Create a separate relation where the PK will contain all the participating entities' keys. - All keys should be underlined and FKs point toward PK.

Superclass / Subclass schema mapping:

Multiple Relations	8A: Multiple relations - superclass and subclasses 8B: Multiple relations - subclass Only
Single Relation	8C: Single relations - with one type attribute 8D: Single relations - with multiple type attribute (flags for each subclass)

8A: Can be applied **for all types** (total or optional, disjoint or overlapping)

8B: Can be applied if the specialization is **Total Only**

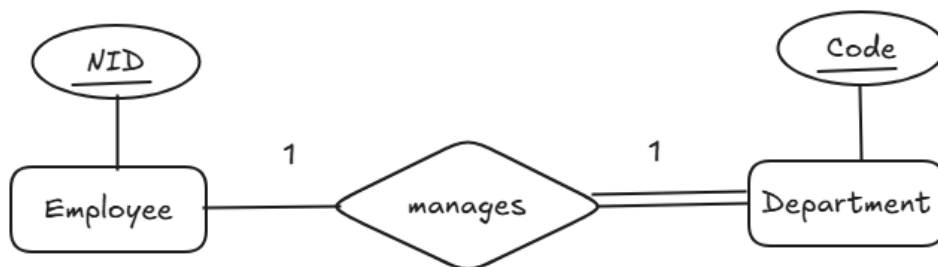
8C: Can be applied if the specialization is **Disjoint Only**

8D: Can be applied **for all types** (total or optional, disjoint or overlapping)

SCHEMA MAPPING EXAMPLES [Pictures Attached]

Binary Relationships

- 1:1 Relationship Mapping

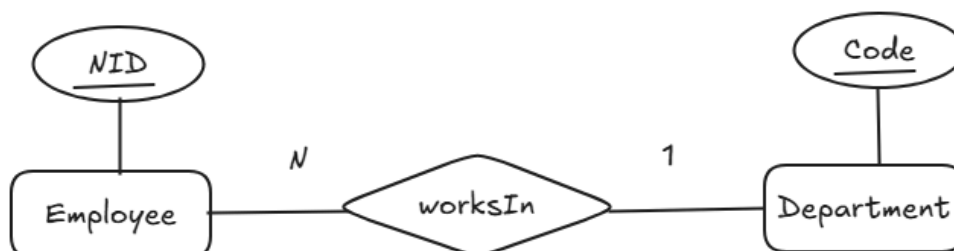


Schema Diagram



Note:
FK approach,
FK put in total side,
NID not underlined in Department

- 1:N Relationship Mapping

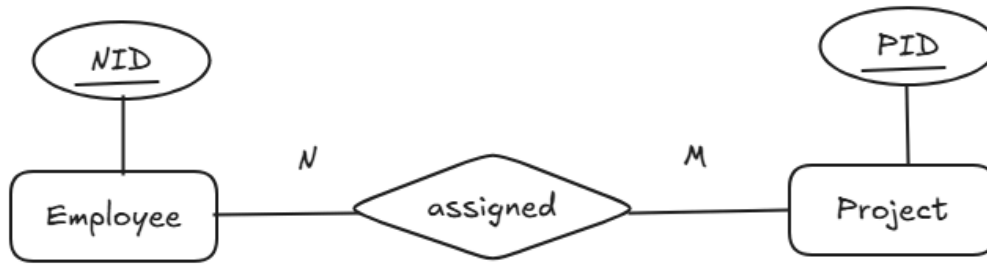


Schema Diagram

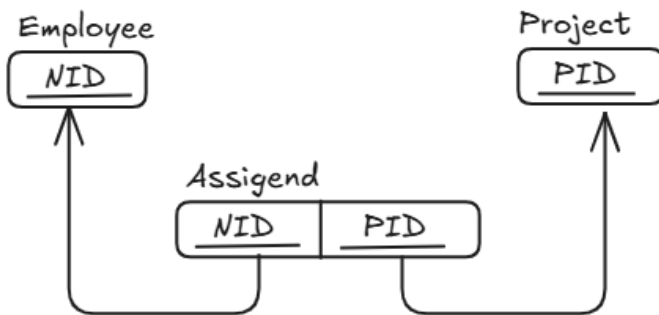


Note:
FK approach,
FK put in N side,
Code not underlined in Employee

- M:N Relationship Mapping



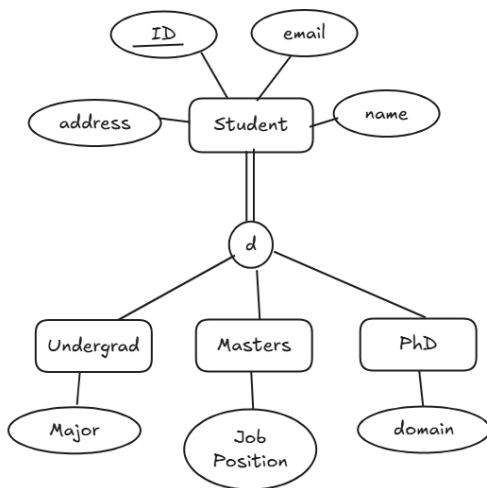
Schema Diagram



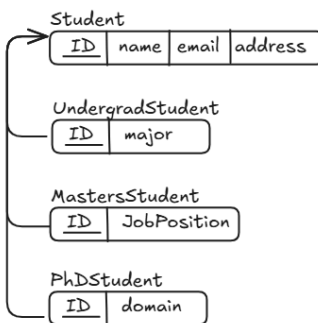
Note:

Cross Referenced Approach,
Create Separate Relation,
Both FK are underlined

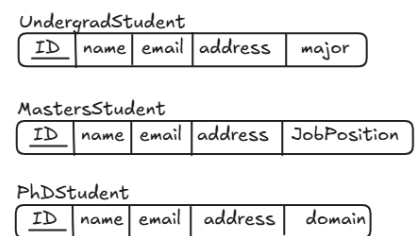
Specialization/Generalization Mapping



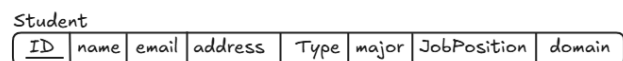
8A Mapping:



8B Mapping:



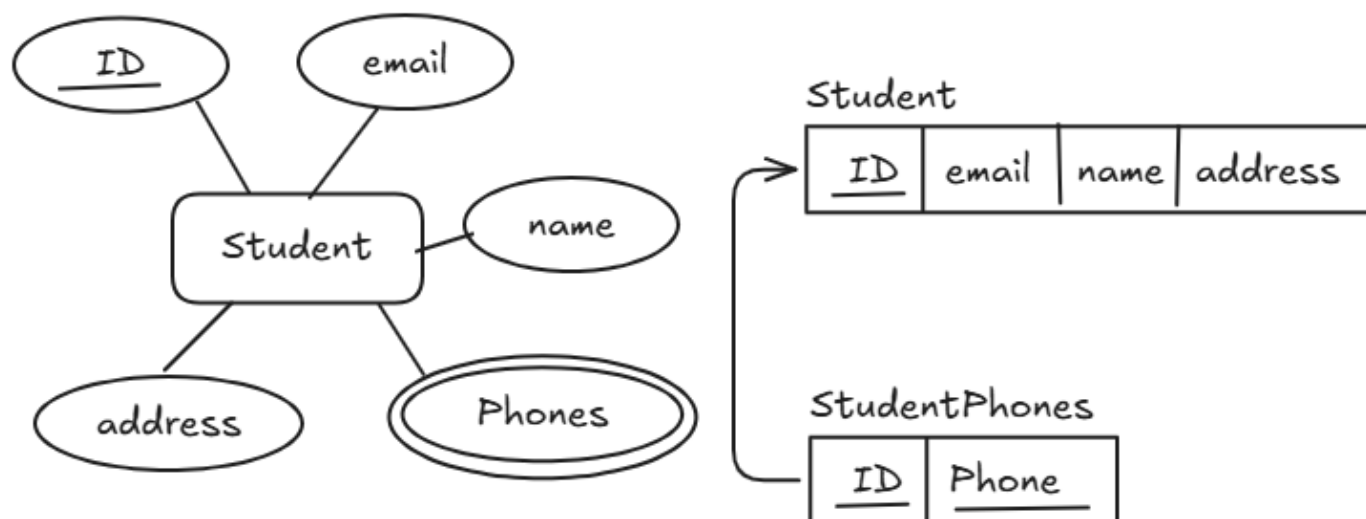
8C Mapping:



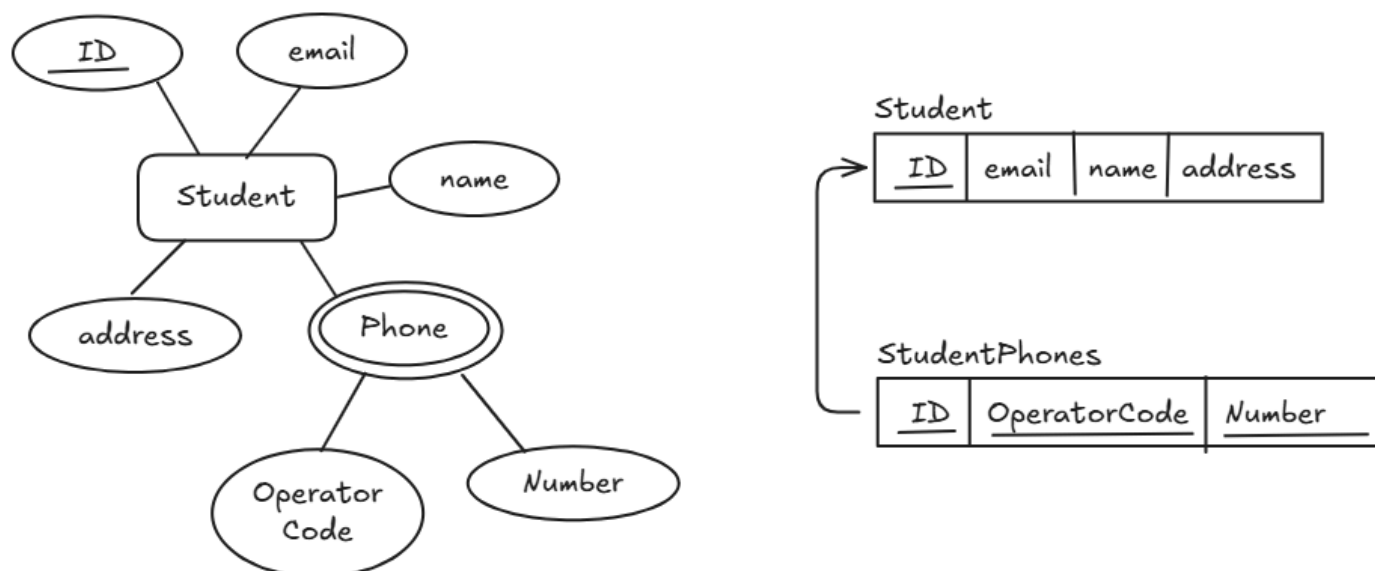
8D Mapping:



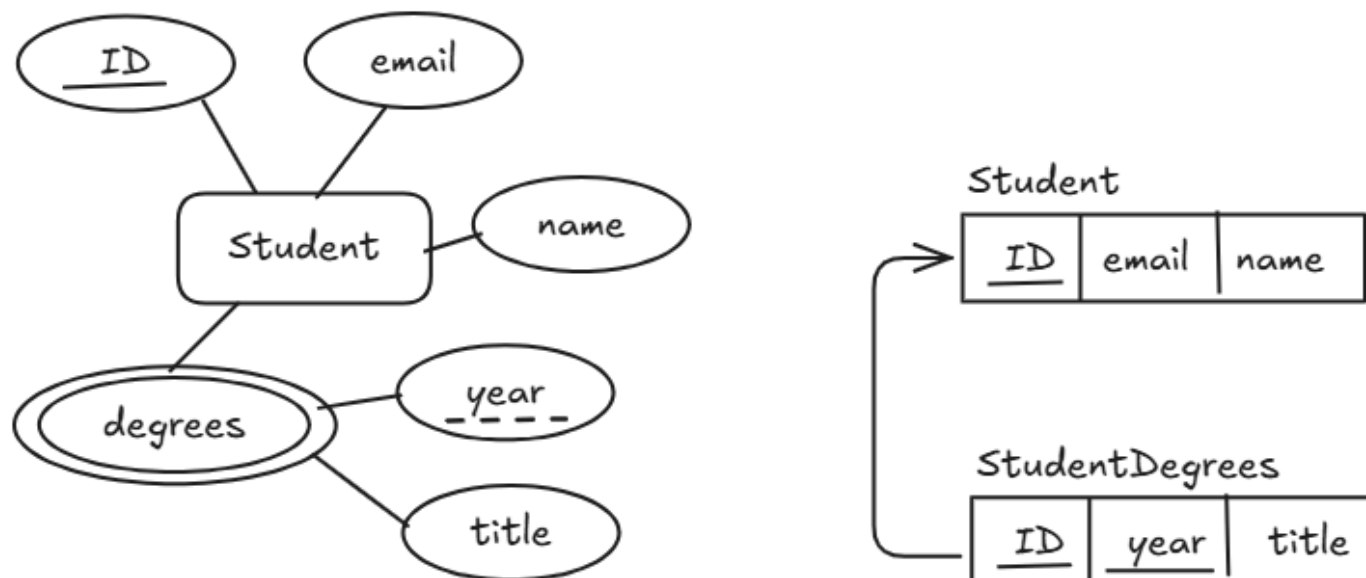
- Multivalued Attribute



- Multivalued Composite Attribute



- Multivalued Composite Attribute with Partial attribute



- Recursive Relationship

